MARKING SCHEME - Generative AI & ML Fundamentals - Mid-Term (Sample)

General: Award marks for correct ideas expressed in any wording. Do not deduct for spelling or grammar. For numerical questions, award method marks even if the final value is wrong (carry-forward errors are not penalised twice).

Question 1 [4 marks]
* Defines overfitting: the model fits the training data (including noise) too closely, so it performs well on training data but poorly on unseen/test data. - 2 marks
* Names a valid technique (e.g. regularisation, dropout, early stopping, more data/augmentation, simpler model, cross-validation) and explains how it helps. - 1 mark
* Names a second, different valid technique and explains how it helps. - 1 mark

Question 2 [6 marks]
* Explains that each token is projected into query, key and value vectors using learned weight matrices. - 1.5 marks
* Attention scores are dot products of a token's query with all keys (QK^T). - 1 mark
* Scores are normalised with softmax to give attention weights that sum to 1. - 1 mark
* Output for each token is the attention-weighted sum of the value vectors. - 1 mark
* Explains scaling by sqrt(d_k): large dot products in high dimensions push softmax into saturated regions with tiny gradients; scaling keeps the variance stable. - 1.5 marks

Question 3 [5 marks]
* Precision = TP/(TP+FP) = 40/50 = 0.8 (formula with correct substitution). - 1 mark
* Recall = TP/(TP+FN) = 40/60 ~ 0.667. - 1 mark
* F1 = 2PR/(P+R) correctly applied ~ 0.727 (accept carry-forward from the student's own P and R). - 2 marks
* Accuracy = (TP+TN)/total = 70/100 = 0.7. - 1 mark

Question 4 [7 marks]
* GAN training: a generator and discriminator are trained adversarially (minimax game); generator tries to fool the discriminator. - 1.5 marks
* Diffusion training: data is gradually corrupted with Gaussian noise (forward process) and a network learns to reverse it / predict the noise (denoising objective). - 1.5 marks
* Stability: GAN training is unstable/hard to balance; diffusion training is stable (simple regression-style loss). - 1.5 marks
* Diversity: GANs can suffer mode collapse (low diversity); diffusion models cover the data distribution better / more diverse samples. - 1.5 marks
* Speed: GANs generate in a single forward pass (fast); diffusion needs many iterative denoising steps (slow), though distillation/fewer-step samplers help. - 1 mark

Question 5 [8 marks]
* Indexing: documents are split into chunks, embedded, and stored in a vector index/database. - 1.5 marks
* Retrieval: the query is embedded and the most similar chunks are retrieved (similarity search, top-k). - 1.5 marks
* Generation: retrieved chunks are inserted into the LLM prompt as context and the LLM generates an answer grounded in them. - 1 mark
* First valid failure mode clearly identified with a specific, appropriate mitigation (e.g. irrelevant/missed retrieval -> hybrid search, re-ranking, better chunking). - 2 marks
* Second, different valid failure mode with a specific, appropriate mitigation (e.g. model ignores context or hallucinates -> instruct to cite sources/abstain, faithfulness checks; stale index -> re-indexing). - 2 marks